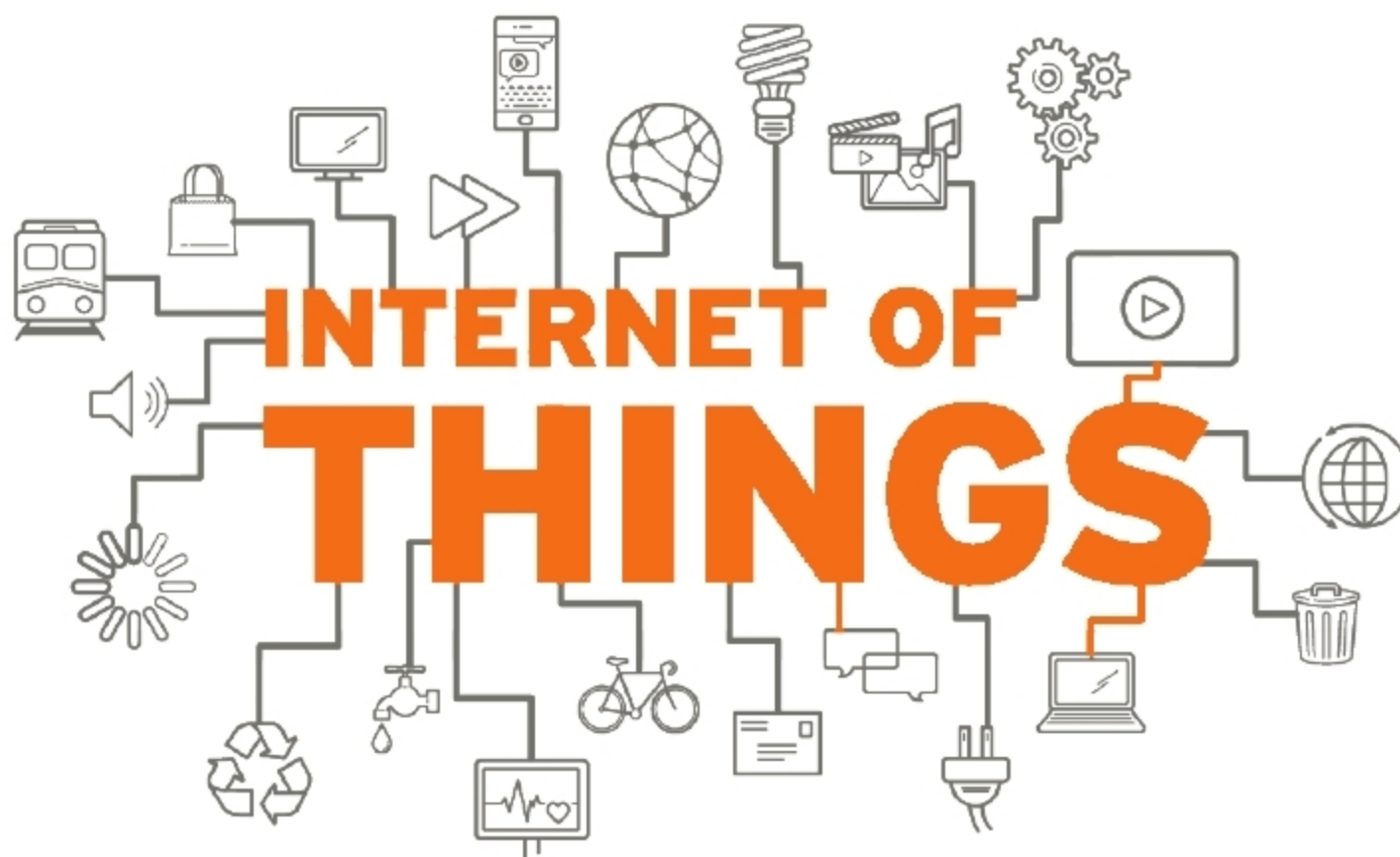


The Benefits *of* Open Source Architecture for IoT

As long as there are developers working in both the proprietary software and the open source software spaces, the debate about which one is better will continue. Proprietary software is driven by the profit motive whereas open source is driven by the concept of sharing knowledge. In this article, the authors discuss the advantages of using open source architecture in the IoT.



Imagine a world where billions of IP-connected objects are sensing, communicating and sharing information. Imagine these objects regularly collecting data, analysing it and initiating action – unleashing a new wealth of intelligence for planning, management and decision making. This concept is called the Internet of Things (IoT).

The IoT refers to everyday objects that are readable, recognisable, locatable, addressable and/or controllable through the Internet, irrespective of the means of communication (whether RFID, wireless LAN, wide area networks, or others).

Examples of ‘things’ include:

- People
- Locations (of objects)
- Time information (of objects)
- Conditions (of objects)

These ‘things’ of the real world seamlessly integrate into the virtual world, enabling anytime, anywhere connectivity.

The impact of the IoT is being felt across a wide range of industries and applications, including agriculture, medicine, manufacturing, electronics, consumer durables, transportation and energy. Looking at the unprecedented number of

connected devices, the pace of change, the rapid innovations, and the wide variety of standards and use cases that the IoT enables, organisations are increasingly looking at open source architectures and technologies. These will drive innovation and accelerate development. The IoT technology ecosystem is typically complex and includes hardware, connectivity, platforms, middleware, analytics, applications and services.

The seven high-level architecture requirements for an IoT based system are:

1. **Context:** Captures the context at all times, 24 hours a day, 365 days a year
2. **Standards:** Uses standards based communication protocols between IoT devices and enterprise systems
3. **Scalability:** Responds to increased load by a decline in performance, not failure; increases capacity proportionally as resources are added
4. **Data management:** Efficiently manages huge volumes of data
5. **Connectivity:** Provides high network connectivity for large data payloads and continuous streaming
6. **Security:** Moves and encrypts information securely even as IoT introduces new risks and vulnerabilities